



Baja.bio

The AI-powered genome browser

One workspace for genomics, oligo design, RNA prediction, literature, and intellectual-property analysis.

Visualize a gene, inspect clinically relevant variation, design a nucleic-acid therapeutic or assay, predict its behavior, and evaluate the evidence around it - without moving between disconnected tools.

Genome browser & visualization

- ✓ Multi-track browser - editable tracks with live parent/child synchronization
- ✓ Variants in context - SNPs, indels, and mutations across the selected region
- ✓ Trusted sources - ClinVar, gnomAD, dbSNP, and COSMIC
- ✓ Variant navigator - filter by source, type, location, and clinical significance
- ✓ Transcript building - collapse exons into a spliced mRNA

Oligo & assay design

- ✓ Ultra-fast GPU off-target analysis - transcriptome-wide screening for siRNA and single-stranded ASOs
- ✓ ASOs and gapmers - chemistry-aware wing/DNA-gap/wing patterns
- ✓ qPCR primers and probes - primer3 designs ranked by expected success
- ✓ Design-by-rules - add and tile oligos across a target with live feedback
- ✓ Structure-aware design - fold RNA and design around stem-loops

Data & analysis layers

- ✓ RNA-seq and IP tracks - coverage, peaks, and binding layers
- ✓ Protein and ORFs - translate sequence and work in protein space
- ✓ Sequence tools - annotate, edit, label, and highlight regions
- ✓ Patent and IP mapping - place claims directly onto sequence
- ✓ Document extraction - load genes and mutations from PDF, text, or image

AI-driven prediction engines

- ✓ BajaSplice - predicts splice sites, exon inclusion, competing-site use, and cryptic exons
- ✓ BajaCLIP Predict - maps sequence-intrinsic footprints for 170 RNA-binding proteins
- ✓ djPrimer - ranks primer3-designed qPCR assays by likely validation success
- ✓ Track-level output - model results appear directly on genes, transcripts, and sashimi layers
- ✓ Decision support - focus experiments, shrink screens, and prioritize bench work

FROM SEQUENCE TO DECISION

Built to keep scientific context attached to every design decision.

Three models that turn sequence and expression into practical experimental priorities

BajaSplice, BajaCLIP Predict, and djPrimer run inside the browser and return track-level predictions, ranked candidates, and exportable coordinates. They are research-use decision-support models; experimental confirmation remains essential.

BajaSplice	BajaCLIP Predict	djPrimer
Human splicing prediction. ✓ Four engines: splice sites, PSI, site usage, cryptic exons ✓ Splice sites: 0.935 PR-AUC ✓ Exon inclusion/PSI: 0.965 AUC ✓ Flags 96% of ClinVar pathogenic splice variants ✓ Finds cryptic-exon candidates, including STMN2	RBP-footprint predictor for ASO screens. ✓ 170 RBPs per 64-nt scan; 48 at AUROC ≥ 0.90 ✓ TDP-43: 0.96 held-out AUROC ✓ Trained on 6.9M ENCODE/eCLIP sites plus POSTAR3 ✓ STMN2 screen example: about 5,500 to about 40 ASOs ✓ Outputs hit maps, clusters, TSV/BED, and sensitivity tracks	Expression-aware qPCR prioritization. ✓ Design with primer3; prioritize with djPrimer ✓ Primer3 scores alone predicted outcomes near chance ✓ Measured on 2,800+ primer/probe sets and 100,000+ results ✓ Gene-grouped cross-validation limits information leakage ✓ Bottom-quartile triage avoided about half of failures

WHO IT'S FOR

A serious research environment with a remarkably low barrier to entry

Baja.bio is designed for people who want to move beyond simply reading about genomics and begin exploring real sequences, evidence, therapeutic ideas, and assay designs in one coherent workspace.

Academics

Explore hypotheses, teach from real genomic context, and connect sequence features to literature, variants, and experimental designs.

Biotech students

Learn the actual workflow of therapeutic and assay design - from gene and transcript to oligo, structure, off-targets, and validation.

Independent researchers

Gain an integrated workbench without assembling and maintaining a fragmented stack of specialist tools.

Physicians and clinician-scientists

Investigate clinically relevant variants and molecular mechanisms with visual context and traceable source material.



For the scientifically curious

You do not need to be a career scientist to use Baja.bio. It is also for curious non-scientists ready to take their understanding to the next level - by working directly with the same kinds of sequences, annotations, evidence, and design logic used in modern biotechnology.